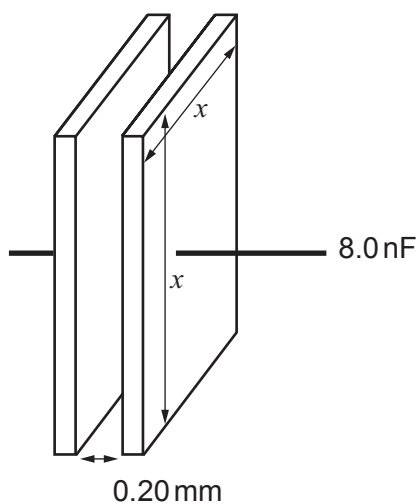


SECTION A*Answer all questions.*

1. (a) (i) An 8.0 nF capacitor has square plates separated by 0.20 mm of air. Calculate the length (x) of the sides of the plates. [3]



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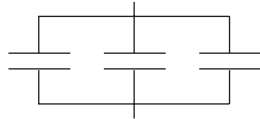


- (ii) Which of the following combinations of 8.0 nF capacitors has a capacitance of 12.0 nF ? Justify your answer. [4]

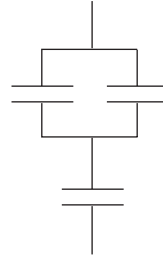
All individual capacitors shown are 8.0 nF .



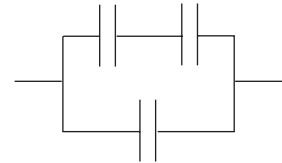
1



2



3



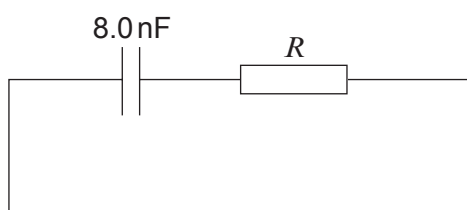
4

- (iii) State how the capacitance of the capacitor can be increased without changing its dimensions. [1]

- (iv) Calculate the charge on the plates of the 8.0 nF capacitor when it stores an energy of $0.62\text{ }\mu\text{J}$. [3]



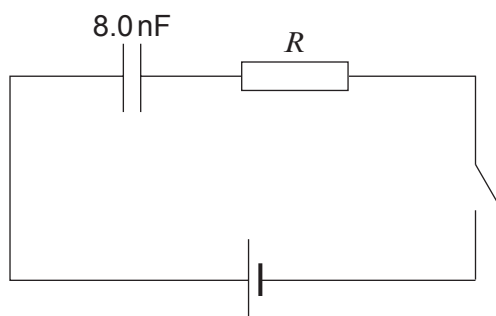
- (b) (i) A charged 8.0 nF capacitor is discharged through a large resistor. Calculate the resistance of the resistor if the time constant is 15.5 ms . [2]



- (ii) Calculate the fraction of the initial charge remaining on the capacitor after 31.0 ms . [3]



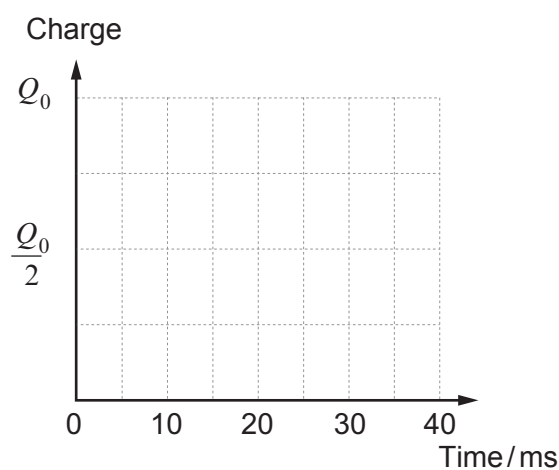
- (c) An identical, uncharged 8.0 nF capacitor is **charged** in the following circuit, **using the same large resistor, R** . The switch is closed and the capacitor is uncharged at time $t = 0$.



Without further calculations, sketch graphs of:

- (i) the charge on the plates of the capacitor against time;

[2]



- (ii) the current in the circuit against time.

[2]

